RAJALAKSHMI ENGINEERING COLLEGE

Janarthan R 250701261-2025-CSE

J2

Started on	Sunday, 30 August 2026, 8:10 PM
State	Finished
Completed on	Sunday, 30 August 2026, 8:18 PM
Time taken	7 mins 55 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Given an array of N integer, we have to maximize the sum of arr[i] * i, where i is the index of the element (i = 0, 1, 2, ..., N).Write an algorithm based on Greedy technique with a Complexity O(nlogn).

Input Format:

First line specifies the number of elements-n

The next n lines contain the array elements.

Output Format:

Maximum Array Sum to be printed.

Sample Input:

5

2 5 3 4 0

Sample output:

40

Answer: (penalty regime: 0 %)

```c
#include<stdio.h>

void swap(int *a,int *b){
    int temp = *a;
    *a = *b;
    *b = temp;
}

int partition(int arr[],int low,int high){
    int pivot = arr[high];
    int i = low - 1;

    for(int j = low; j <= high; ++j){
        if(arr[j] < pivot){
            ++i;
            swap(&arr[i],&arr[j]);
        }
    }

    swap(&arr[high],&arr[i+1]);
    return i + 1;
}

void QuickSort(int arr[],int low,int high){
    if(low <= high){
        int p = partition(arr,low,high);
        QuickSort(arr,low,p - 1);
        QuickSort(arr,p + 1,high);
    }
}

int main(){
    int n;
    scanf("%d",&n);

    int arr[n];
    for(int i=0;i<n;++i)
        scanf("%d",&arr[i]);

    QuickSort(arr,0,n-1);

    int ans = 0;
    for(int i=0;i<n;++i)
        ans += (arr[i] * i);

    printf("%d\n",ans);
}
```

	Input	Expected	Got	
✔	5 2 5 3 4 0	40	40	✔
✔	10 2 2 2 4 4 3 3 5 5 5	191	191	✔
✔	2 45 3	45	45	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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